

Question Paper

Exam Date & Time: 19-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Communication Theory [ECM_3121]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain what is a random process. Discuss the requirements for the stationarity to the first and second order. (4)
- A) Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) Prove the relation between variance of a random variable with its first and second moments as follows: $\text{Var}[X] = E[X^2] - (E[X])^2$ (4)
- Course Outcome: "CO1" / Cognitive Level: "Analysis"
- C) Explain the concept of moments and state the mathematical description of the n^{th} moment. (2)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Identify and briefly describe the blocks in a digital communication system that (4)
- A) i. introduces errors into the signal
ii. adds redundancy
- State one example each of an analog and digital communication system.
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- B) Consider the phase modulated signal $s(t) = 10 \cos(2\pi \cdot 10^7 t + 5 \sin(2\pi \cdot 200 t))$. (4)
- Determine what is the maximum phase deviation and state when it occurs.
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- C) Explain the concept of phase modulation with a mathematical description. (2)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) 1. Explain the concept of equalization. Explain why we need it. (4)
2. Determine the equalizer to achieve distortionless transmission for signal with transfer function: $H_c(f) = 0.5 \exp(-j2\pi f t_c)$.
- A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) Given $x(t) = \cos(100\pi t) + \cos(650\pi t)$. (3)

- I. Determine the highest frequency content.
- II. Calculate the Nyquist rate.
- III. If sampled at $f_s = 500$ Hz, identify and illustrate observed frequencies due to aliasing.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- C) Represent bitstream 01011001 by NRZ-M, Bipolar RZ, and Manchester. Describe the requirements for each scheme in a line or two. (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 4) Explain the concept of information metric $I(x)$. State its mathematical expression and any two of its properties. (2)

A)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- B) Consider the discrete memoryless source with source alphabet $A = \{s_0, s_1, s_2\}$ with their corresponding probabilities $p_0, p_1, p_2 = \{\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\}$. Calculate the entropy. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) Compute the Huffman code for symbol set with probabilities $\{s_0, s_1, s_2, s_3, s_4\} = \{0.4, 0.2, 0.2, 0.1, 0.1\}$. Compute the average codeword length and entropy. In the combined symbol, make moves as high as possible. Repeat by moving as low as possible. Compute variance for both approaches and compare (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) i. Explain the concept of syndrome decoding and describe its working principle. (5)
ii. Describe any two properties of syndrome decoding.
A) iii. List any 3 types of cyclic codes.

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) For a (7,4) cyclic code, the polynomial $(1+D^7)$ is factorized as $(1+D)(1+D^2+D^3)(1+D+D^3)$. Determine the parity check polynomial for the generator polynomial $(1+D+D^3)$. Compute the message and codeword polynomial, for the message sequence (1001). (5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 ECM_3121:Communication Theory

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025
ECM_3121:Communication
Theory

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	2(4%)
3	Application	4	12(24%)
4	Analysis	9	36(72%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	14	50 (100%)